



Topology Optimization for Inspectability

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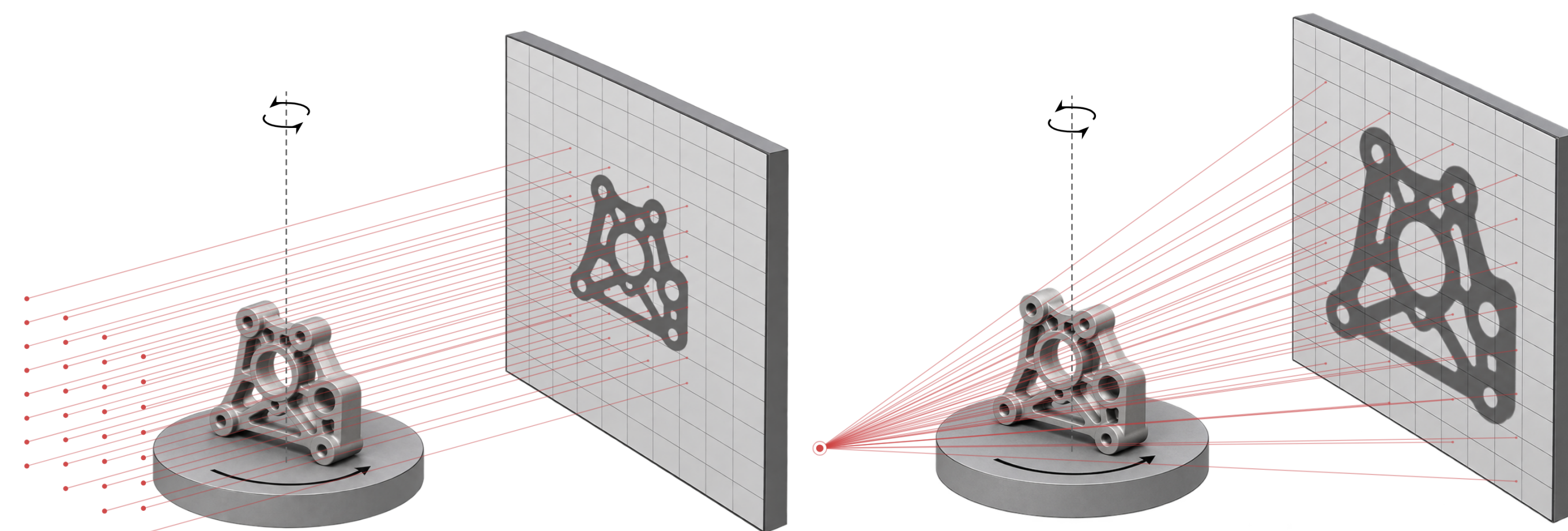
Background

Manufacturable does not mean inspectable. X-ray Non-Destructive Evaluation (NDE) verifies the interior of a part without damaging it. Thick load paths can attenuate too much X-ray intensity for reliable CT inspection. We include inspectability directly in topology optimization by limiting material thickness along inspection paths.

The Beer–Lambert law relates material thickness to detector intensity:

$$I = I_0 e^{-\mu x}$$

For one material, a minimum detector intensity sets a maximum path thickness.



Parallel beam

Cone beam

Optimization formulation

$$\begin{aligned} \min_{\rho} \quad & C(\rho) \\ \text{s.t.} \quad & K(\tilde{\rho}) \mathbf{u} = \mathbf{f}, \\ & g_v(\tilde{\rho}) \leq 0, \\ & g_{A,k}(\tilde{\rho}) \leq 0, \quad k = 1, \dots, N_A, \\ & 0 \leq \rho \leq 1. \end{aligned}$$

$C = \mathbf{f}^T \mathbf{u}$ compliance
 $K(\tilde{\rho}) \mathbf{u} = \mathbf{f}$ elastic equilibrium
 g_v volume fraction
 $g_{A,k}$ attenuation bound, direction k

A Helmholtz PDE filter maps design density ρ to filtered density $\tilde{\rho}$, which enters both the elastic stiffness and attenuation constraints. $\mathbf{u}$ is displacement and $\mathbf{f}$ is the applied load.

Local constraint with slack variable

p -norm surrogate

$$g_{A,k} = \left(\int_{\Omega} a^p d\Omega \right)^{1/p} - \alpha_{\max}$$

Bounded slack field

$$g_{A,k} = \int_{\Omega} (a - \alpha)^2 d\Omega - \epsilon$$

$$\alpha_{\min} \leq \alpha(\mathbf{x}) \leq \alpha_{\max}$$

The p -norm surrogate can permit local bound violations. A pointwise-bounded slack field α tracks thickness a ; continuation reduces the mismatch tolerance as $\epsilon \rightarrow 0$. Finite ϵ allows a residual mismatch.

Evaluating the thickness

Ray tracing

$$a(\mathbf{r}) = \int_0^L \tilde{\rho}(\mathbf{x}(s)) ds$$

Advection model

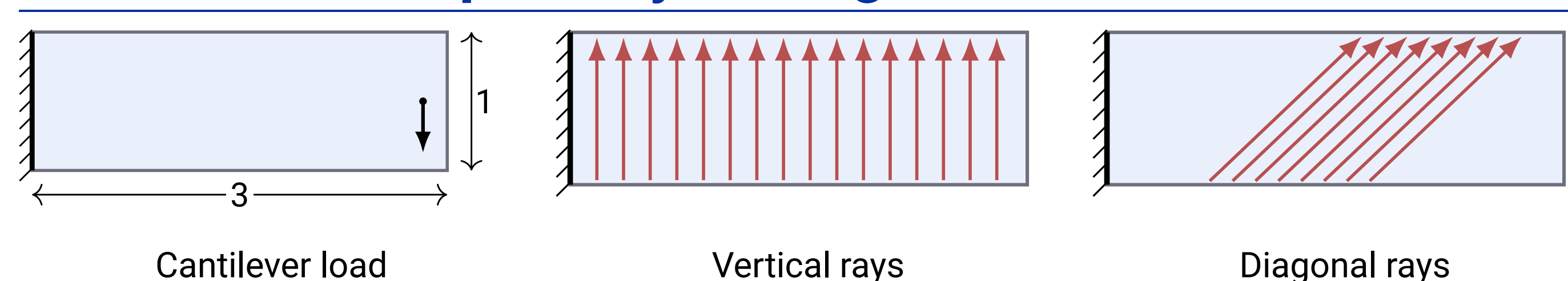
$$\mathbf{v} \cdot \nabla \rho_a = \tilde{\rho}$$

$$\rho_a = 0 \text{ at inflow } \Gamma_-$$

$$\mathbf{a} = \rho_a|_{\Gamma_+}$$

MFEM implementation. Ray tracing uses FindPointsGSLIB to locate sample points and interpolate density. The advection model uses DG with standard MFEM operators and reads thickness at the outflow boundary.

Compare ray tracing and advection

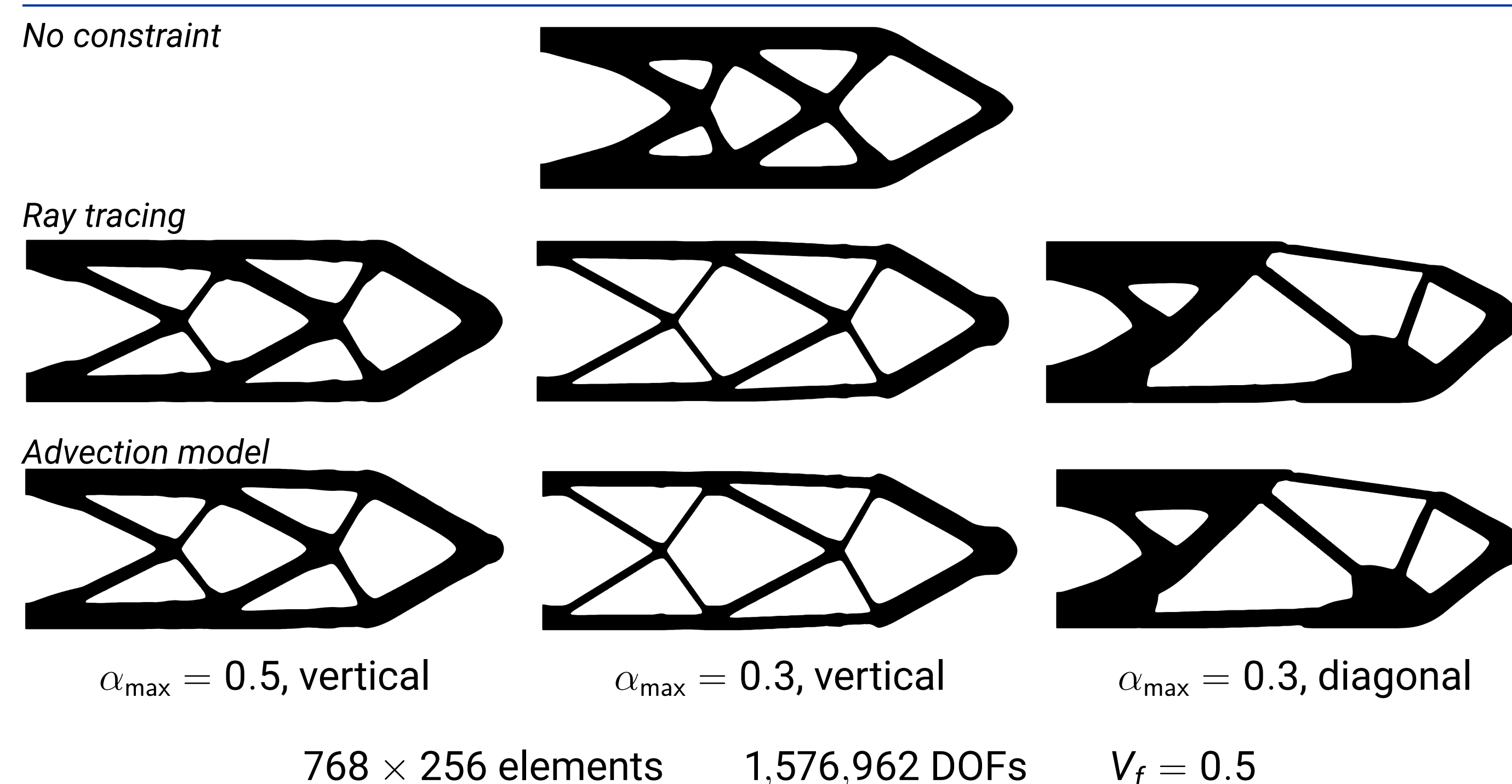


Cantilever load

Vertical rays

Diagonal rays

Compare ray tracing and advection (cont.)



$\alpha_{\max} = 0.5$, vertical

$\alpha_{\max} = 0.3$, vertical

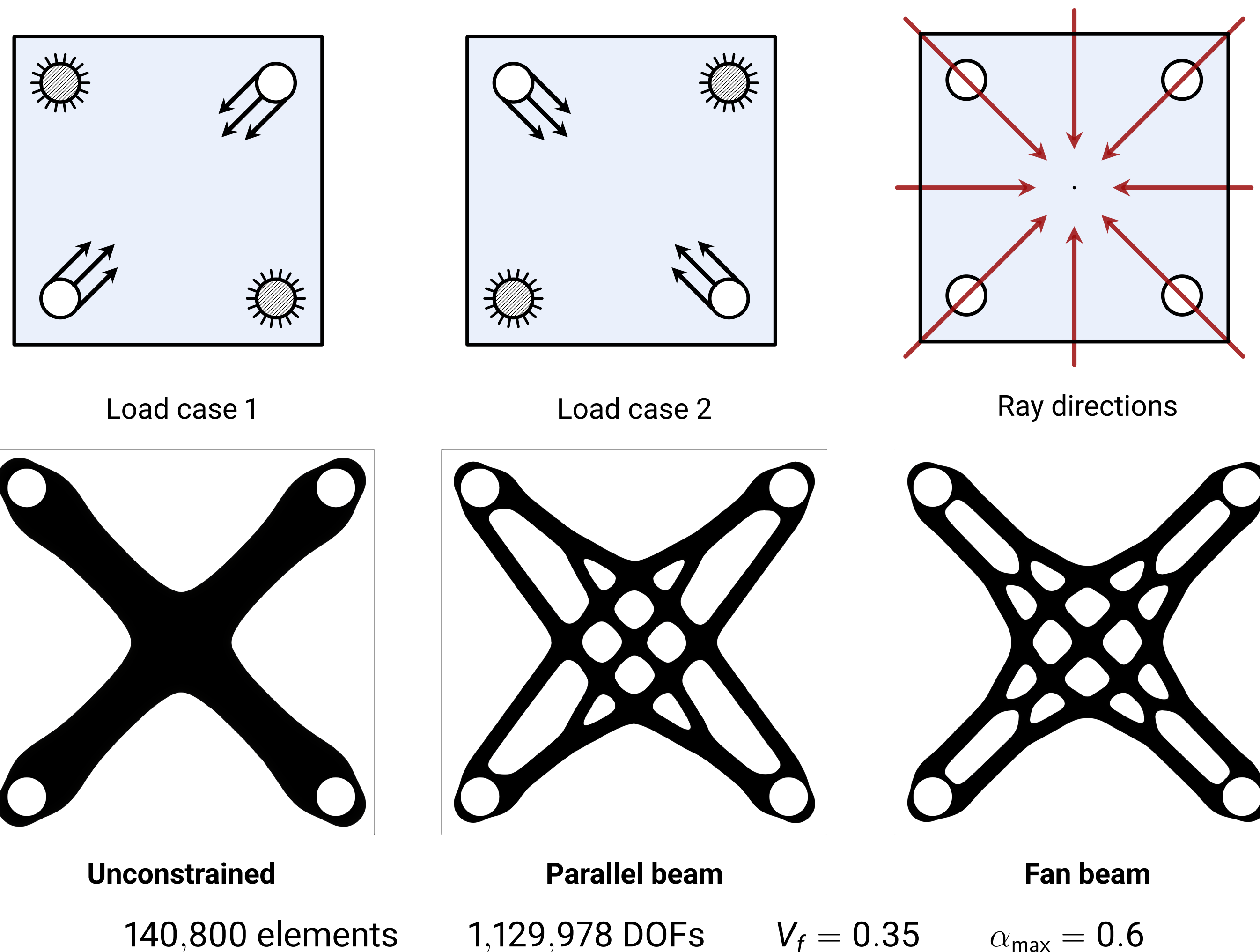
$\alpha_{\max} = 0.3$, diagonal

768 × 256 elements

1,576,962 DOFs

$V_f = 0.5$

Ray geometry changes the design



Load case 1

Load case 2

Ray directions

Unconstrained

Parallel beam

Fan beam

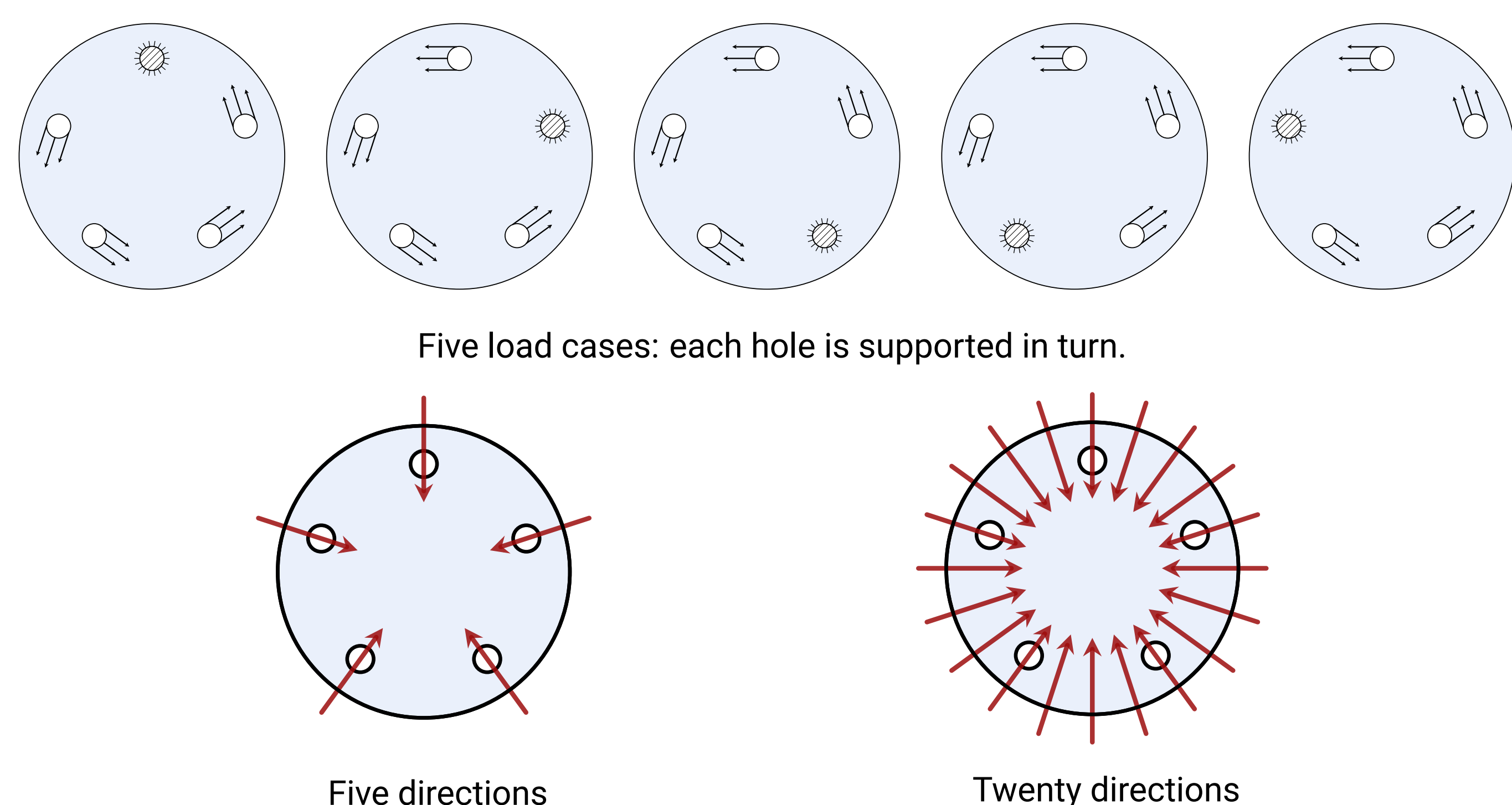
140,800 elements

1,129,978 DOFs

$V_f = 0.35$

$\alpha_{\max} = 0.6$

Vary the directions and thickness bound

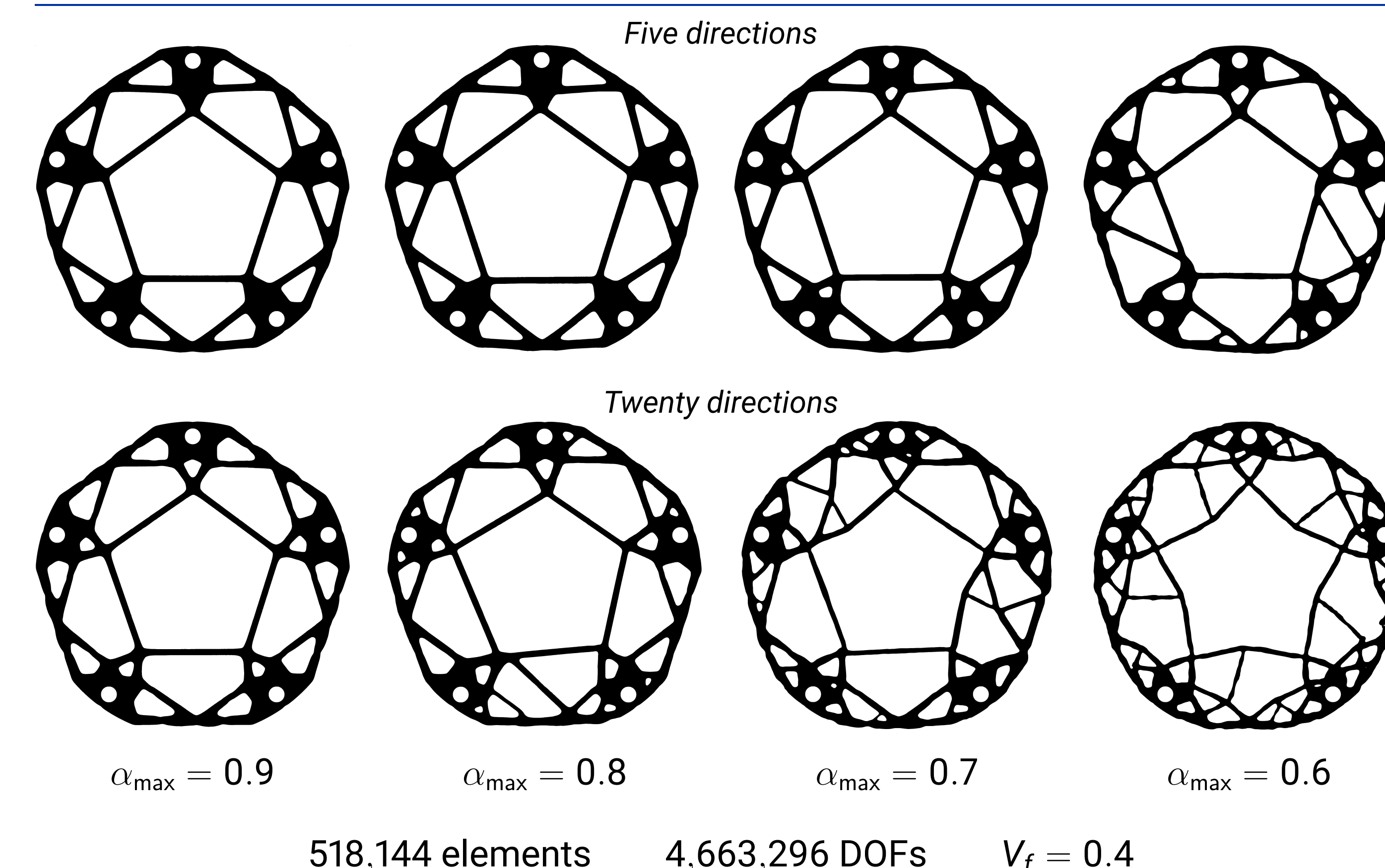


Five load cases: each hole is supported in turn.

Five directions

Twenty directions

Vary the directions and thickness bound (cont.)



Five directions

Twenty directions

$\alpha_{\max} = 0.9$

$\alpha_{\max} = 0.8$

$\alpha_{\max} = 0.7$

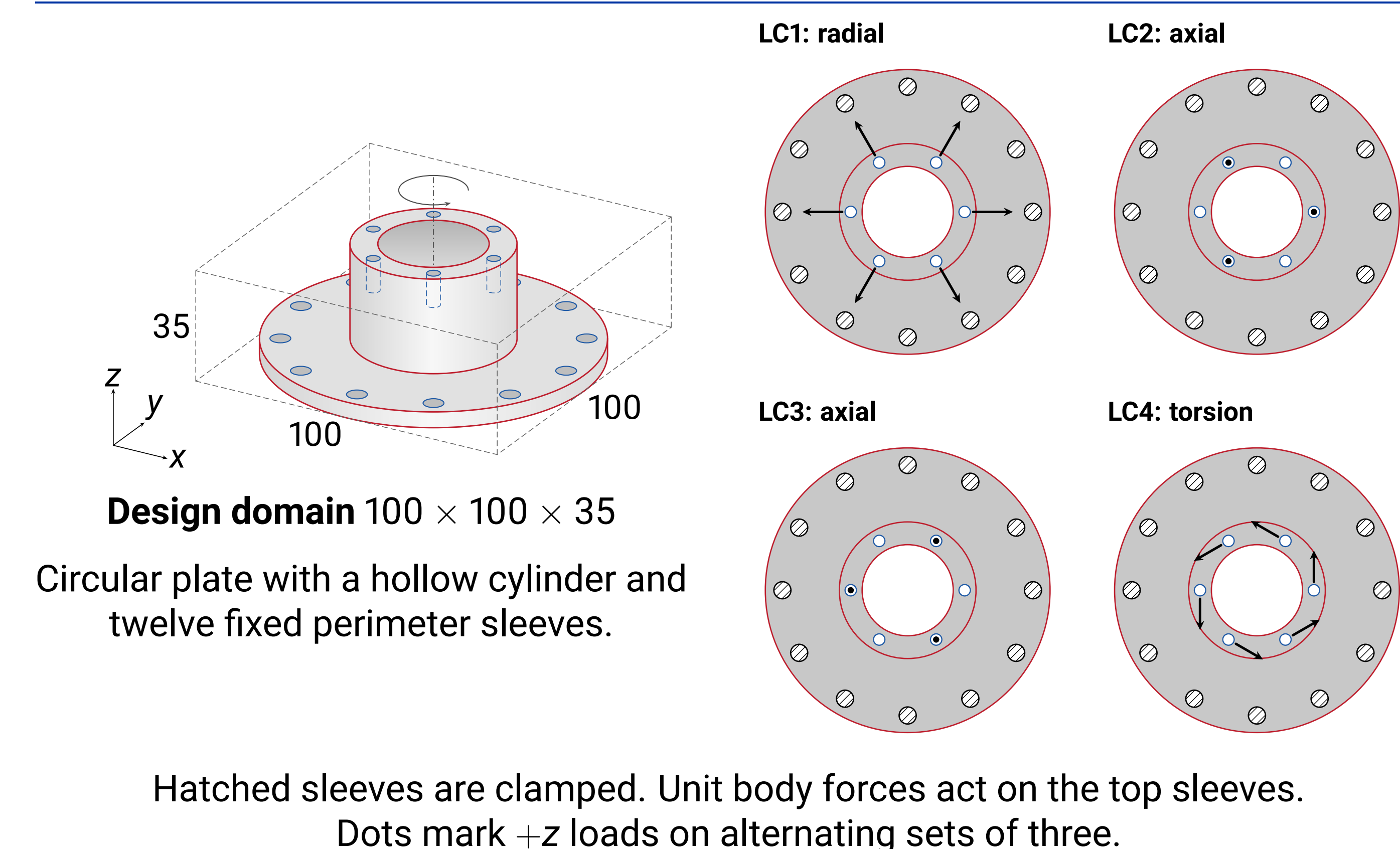
$\alpha_{\max} = 0.6$

518,144 elements

4,663,296 DOFs

$V_f = 0.4$

3D large-scale simulation



LC1: radial

LC2: axial

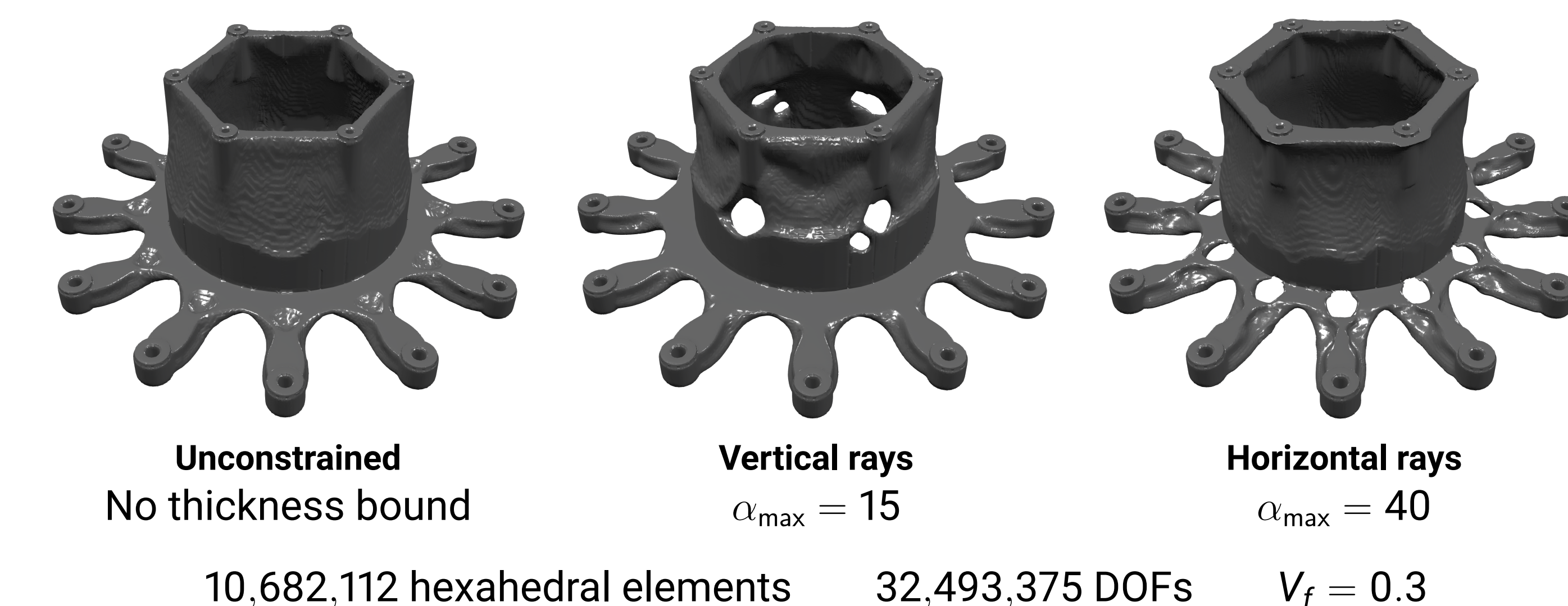
LC3: axial

LC4: torsion

Design domain 100 × 100 × 35

Circular plate with a hollow cylinder and twelve fixed perimeter sleeves.

Hatched sleeves are clamped. Unit body forces act on the top sleeves. Dots mark +z loads on alternating sets of three.



Unconstrained

Vertical rays

Horizontal rays

No thickness bound

$\alpha_{\max} = 15$

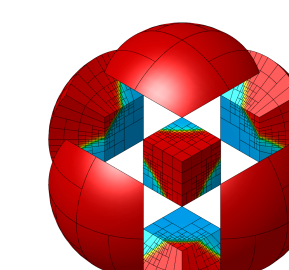
$\alpha_{\max} = 40$

10,682,112 hexahedral elements

32,493,375 DOFs

$V_f = 0.3$

Acknowledgements



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